Class 1.5 — Control Box I

Sunday, July 12, 2026 · 3 hours · Mr. Holcombe's house

THE ONE GOAL

Every kid earns a crafted joint on the PufferFish and contributes at least one joint to the TriggerFish board. Working-vs-crafted is the frame they leave with.

BEFORE YOU TEACH THIS

- Watch the MATE "How to Solder Components to a PCB" presentation end-to-end (15 min). Even if you've soldered for years, watch it — the kids will recognize lines from it and you'll match their reference frame.
- Solder a PufferFish board yourself the night before. Time yourself. If it takes you 45 minutes, expect the kids to take 60–75.
- Read pages 377–388 (§8.4) and 392–434 (§8.6–8.7) of the Underwater Robotics textbook (Preston's copy) — the electronics chapter that the SeaMATE Guide tracks against.
- Open the SeaMATE TriggerFish ROV Guide and find the exact slide where 1.3 left off. Bookmark it; don't search for it during class.
- Confirm the 12V SLA battery + charger order has shipped or is in hand — it's the gating item for 1.6. Without it, the smoke test slips a week.
- Internalize Section 4.3: every joint on every board is student work. You teach the move once on the PufferFish; after that, you photograph and you watch.

WHY THIS CLASS MATTERS

1.5 is the bridge class. Up through 1.4 we taught skills in isolation — soldering, multimeter, simple circuits. Starting tonight those skills land on the real ROV. The first time a kid puts a hot iron on the TriggerFish board, the season changes shape.

The frame we want the kids to leave with is "working vs. crafted." A joint that passes a multimeter check might still fail under vibration or in the water. A crafted joint — one that's flush, shiny, meter-confirmed, and built to a checklist — doesn't. The PufferFish is where we calibrate what crafted feels like before the stakes go up.

HOW THE MEETING RUNS — BLOCK BY BLOCK

0:00 – 0:15 · Welcome and the safety brief

Start on time. Read the agenda end-to-end so the kids have the shape of the night, then hand the standing safety brief to the assigned Safety Briefer if they're ready, or run it yourself if not. The four standing questions every meeting, then today's brief card. Slow down on prevention — the back-of-board heat hazard catches new solderers off guard because they're watching the iron tip, not the leads.

Hand off the four rotating roles right here: Notetaker, Safety Briefer, Photographer, Timekeeper. Names on the slide. The Notetaker carries the action-items slide at the wrap-up; they need to know they're on the clock for the whole meeting.

0:15 – 0:30 · The PufferFish — practice board brief

Frame the why before the how. The PufferFish has nothing to do with the ROV; that's exactly the point. It's a low-stakes circuit where a botched joint costs nothing and a clean joint earns the right to touch the TriggerFish. Use the words "working vs. crafted" out loud and define both — a multimeter-passing joint is working; one that also survives heat, vibration, and the trip to the pool is crafted.

Walk the five moves on the slide. Demonstrate one joint at the bench while the kids watch — that's the one demonstration. After that, every joint is theirs. Section 4.3 starts here.

0:30 – 1:15 · PufferFish — hands on

Forty-five minutes, the kids at the bench. Your job is to photograph, observe, and ask one diagnostic question per kid when they're unsure. Do not pick up an iron. If a kid asks how to do it, point them back to a teammate who just did it correctly — that's the mentor pattern.

Walk the room. Look for cold joints first (dull or grainy surface) — these are the ones the multimeter will pass but the field won't. Quietly point them out and watch the kid re-flow. By minute thirty, every kid should have at least three crafted joints behind them.

1:15 – 1:30 · The TriggerFish REV 4 — where we pick up

iPad on the bench, SeaMATE Guide on the bookmarked slide. Read what was done in 1.3 — fuse holder, two 1000µF / 16V caps, 3 A fuse seated, power leads heatshrunk — then read the next block of components from the Guide as a team. One kid reads, the others listen. No iron picked up yet; this is the reading block.

Reinforce the polarity rule: every electrolytic cap, diode, and transistor gets its orientation verified against three references — the silkscreen, the Guide image, and the component itself — before any commit. A reversed cap on this board fails loud and ends the night.

1:30 – 1:40 · Break

Ten minutes. Iron tips go in holders, kids step away from the bench. Resume at 1:40 — start on time, the habit is part of the engineering.

1:40 – 2:30 · TriggerFish — next block of components

Fifty minutes, hands on. The build loop is on the slide: read the slide together, stage the components, solder one student at a time, inspect and meter-check, move to the next slide only when the current passes. Hold that rhythm — do not let the team get ahead of the Guide.

If a joint fails inspection or meter-check, the joint gets re-flowed before the next move. We never "fix it later." Later means at the regional, in the hotel parking lot, with the multimeter dying and the deck call in fifteen minutes. We fix it now.

2:30 – 2:45 · Meter-check discipline

Bring the room back together. Walk the three reads on the slide — continuity, no shorts, polarity confirmed — and explicitly tie the discipline to a competition outcome: at safety inspection, the judges will probe the board the same way. The team that's been metering every stage all season passes inspection clean. The team that didn't, doesn't.

Have a student run the three reads on one joint at the bench for the room to watch. That demonstration counts as their crafted-joint receipt for the night.

2:45 – 3:00 · Wrap-up and homework

Read the three student items out loud, confirm each has an owner. Hand off to the Notetaker for the action-items slide; they read back what they captured tonight, and you confirm each item before it leaves the room.

Close on the parallel-track card — the 12V SLA battery and charger are the gating item for 1.6, and the parents need to hear that out loud. Then the tool-checkout slide; the Tool Steward leads it. Doors at 5:00; closer slide carries them out.

> **THE PARENT LANE — SECTION 4.3**
> The kids solder; you photograph. That's the rule for tonight. You demonstrate the five moves once on the PufferFish — that's it. Every joint after that is a kid's. If a kid is stuck, your move is to ask one diagnostic question ("is the pad hot before you fed the solder?") and step away. You do not pick up the iron. The judges in April will ask each student to explain how a joint was made; the team that hears "I made this one" gets the points. The team that hears "my coach helped me with that part" does not.

COMMON PROBLEMS & QUICK FIXES

Symptom	Quick fix
Cold joint — dull or grainy surface	Re-flow. Heat the pad and the lead together for a full second before feeding solder. Cold joints almost always mean the iron wasn't on the pad long enough.
Solder bridge between two pads	Add flux, re-heat both pads, wick away the excess with desoldering braid. Then meter-check for shorts before moving on.
Joint won't take solder no matter how long the iron sits	Iron tip is oxidized — wipe on the brass cleaner, re-tin with fresh solder, try again. If still failing, the iron tip needs replacement.
Component sits crooked on the board	Re-heat one lead while pressing the component flat from the other side. Then re-heat the second lead. Components must seat flush — a crooked component is a stress riser on the joint.
Reversed electrolytic cap discovered after soldering	Stop. Do not power on. Desolder both leads, remove the cap, install a fresh one (the old one is potentially compromised even if it looks fine). Re-verify polarity with team member before committing.
Iron tip won't reach 650°F	Discard the iron. Substitute from the backup. Note it on the BoM so it gets replaced before the next class.

WHAT "DONE" LOOKS LIKE

- Every kid has at least one crafted joint on a PufferFish board they can explain to a parent.
- The TriggerFish board advances the next block of components per the SeaMATE Guide, with every joint meter-checked.
- No reversed polarity-sensitive components on the TriggerFish board.
- Action-items slide is filled in and read back before the closer.
- 12V SLA battery + charger gating item is named out loud, with the parent-side owner confirmed.

IF YOU FALL BEHIND

If the PufferFish block runs long, do NOT skip the meter-check discipline section — that's the section that protects 1.6's smoke test. Cut the TriggerFish block short and pick it up at the start of 1.6. The frame (working vs. crafted) is more important than the slide count.